Nathan Ritchie

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Education

University of Western Ontario **2019 - 2025**
Bachelor of Science: Mechanical and Materials Engineering (Dean's Honour List) 2019 - 2024
Bachelor of Science: Computer Science (Dean's Honour List) 2022 - 2025

Summary of Qualifications

- Experienced in Java, Python, C/C++, Matlab, SQL, JavaScript, and PHP
- Experienced in C and UNIX based systems
- Developed Python scripts and test fixtures at previous internship for quality control processes
- Experienced with real time microcontroller systems
- Experience with electrical design, component selection, and PCB assembly

Academic Projects/Course Work

Smart Home Controller
- Developed a Smart Home Controller in **C/C++** using the **Qt** framework
- Designed a responsive **GUI** using Qt widgets that displays real time smart home information and device states

Heart Disease Prediction using Machine Learning
- Built and optimized machine learning models (Logistic Regression, Random Forest, XGBoost, etc. in **Python**, achieving a AUC-ROC score of 0.948 for heart disease prediction, showcasing strong prediction performance
- Conducted comprehensive data preprocessing (e.g., robust scaling, one-hot encoding) and visualized key trends using Matplotlib, uncovering critical predictors to improve model interpretability and drive decision making.
- Delivered insights through advanced visualization techniques, including scatter plots, correlation heatmaps, and feature importance analyses, enabling data driven strategies for risk assessment and early intervention.

BeStep (Flutter, Dart, Firebase)
- Created a **cross platform** social media alternative to **BeReal**, using **Flutter, Firebase**, and Native **Pedometer Libraries.**
- BeStep is a BeReal alternative where you take and share pictures with friends upon achieving specific step milestones.
- Implemented user movement tracking via native pedometer libraries to calculate steps taken and moving status.
- Architected the apps backend with **firebase** libraries to **authenticate** users, **store** user images, and **log** any crashes.

Desktop Spell Checker (Java, OOP, Design Patterns)
- Lead a 4-Person Agile development team using the scrum framework to create a desktop spell checker application that recommends corrections to files as they are being edited.
- Organized a product backlog and distributed tasks to team members for each code sprint.
- Improved code architecture by using OOP principles and design patterns utilizing Git to work on a shared repository.

Multiplayer Online Snake Game (Networking, Python)
- Developed a multi-player server-client snake game in Python, managing multiple client connections and implementing game logic on the server side.
- Enhanced the game with real-time public messaging feature, allowing clients to broadcast messages to all participants.
- Implemented RSA encryption for secure communication between clients and server, ensuring data integrity and privacy.
- Designed communication protocol to support new game functionalities, including multi-client connectivity and encrypted data exchange.

Scavenger Robot (C++, ESP microcontrollers, autonomous systems, PID control)

- Developed a robot that was remotely operated over a WiFi network tasked with collecting and autonomously distinguishing valuable/waste objects
- Gained experience interfacing with off the shelf CAN and I2C components
- Developed robust and reliable control structures for the reliable operation of autonomous systems

University T.A. Management Website (MySQL, PHP, HTML, CSS, JavaScript)
- Developed a web application for managing a Teaching Assistant database using PHP, MySQL, HTML, and CSS, with features for data sorting, record insertion, and deletion.
- Implemented detailed Teaching Assistant profile views with conditional image rendering and course preference display.
- Engineered relational database functionalities for assigning TAs to courses and built a course offering viewer with customizable date ranges.
- Ensured best practices in code structure, modularity, and documentation for maintainability and readability.

Work Experience

Metro Vancouver Regional District
Engineering Technician — October 2024 - Current
- Developed **PowerBi** dashboards to better communicate energy usage data. 10-15 hours of work saved weekly
- Built **python** data pipelines to automate data refreshes for PowerBi dashboards
- Applied Industry Standard measurement and verification techniques to quantify performance of capital projects

Rocket Lab Space Systems
Business Development Intern — May 2023 - August 2023
- Utilized Python(pandas) to investigate product failure modes and refine safe operational envelope figures
- Automated the development of a product database with python using PDFquery to gather historical data that was not centrally recorded
- Developed preventative maintenance planning and automated work scheduling using Infor LN

General Motors
Manufacturing Process Engineering Intern — May 2022 - December 2022
- Oversaw retooling of entirely new electric vehicle bodyshop. Managed budget and roadblocks
- Implemented plc control algorithms to introduce error proofing to production. Reducing downtime and improving quality
- Managed the deployment of new network architecture to improve plant monitoring and control
- Designed fixtures for factory retooling. Balancing the needs of Production, Management, and Engineering stakeholders
- Developed LEAN manufacturing solutions to deal with damaged parts or downtime scenarios
- Managed a group of 24 tradespeople. Prioritized/Assigned work, Coordinated contractors, and summarized work for next shift

Rocket Lab Space Systems
Engineering Intern — May 2021 - August 2021
- Responsible for the development of laser vibrometry QA test. Involving concept generation, mechanical design, and data processing using Python/Pandas
- Developed product functional testing scripts for thermal vacuum testing
- Mechanical design and implementation of complex qualification testing fixtures. Involving large assemblies and FEA analysis
- Frequently participated in soldering and wiring harness building for various pieces of flight hardware and ground support equipment
- Familiar with operating power supplies, oscilloscopes, multimeters, and other diagnostic equipment
- Helped to improve the design of a new product to improve repeatability in manufacturing and minimize drift from specifications in transport

Other Interests

Mountain Biking • Skiing and Snowboarding • Rock Climbing • Automotive Performance Tuning